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EMPLOYMENT

- 01/2024 – now **Postdoctoral researcher**
Max Planck Institute of Colloids and Interfaces, Germany *Colloid Chemistry Department*
Advisor: Prof. Dr. M. Antonietti.
- Research on biomass conversion into artificial humic substances.
- Supported by an Alexander Von Humboldt fellowship.
- 05/2022 – 12/2023 **Postdoctoral researcher**
University of Trento, Italy *Department of Civil, Environmental and Mechanical Engineering*
Advisor: Prof. L. Fiori.
- Hydrothermal conversion of biomass and bioplastic residues into bio-based products.
- 04/2018 – 11/2018 **Industrial engineer intern**
Dana Corporation, Italia

VISITING RESEARCHER

- 05/2025 – 06/2025 **Northeast Agricultural University, China** (host: prof. F. Yang)
08/2023 – 11/2023 **Tohoku University, Japan** (host: prof. T. Adschiri)
03/2021 – 10/2021 **Cornell University, USA** (host: prof. J. Goldfarb)
09/2017 – 01/2018 **Boston University, USA** (host: prof. J. Goldfarb)

ACADEMIC TEACHING

- 05/2025 – 06/2025 **Lecturer** Northeast Agricultural University, China
01/2019 – 07/2023 **Teaching assistant** University of Trento, Italy

EDUCATION

- 11/2018 – 05/2022 **PhD in Civil, Environmental and Mechanical Engineering** University of Trento, Italy
Major: “*Materials, Chemistry and Energy.*”
Advisors: Prof. L. Fiori, Prof. A. Miotello.
- Thesis title: “Sustainable conversion of biomass wastes via hydrothermal processes: fundamentals and technology.”
- 09/2015 – 03/2018 **Master of Science in Energy Engineering**
University of Trento & Free University of Bozen-Bolzano, Italy
Final grade: 110/110 cum laude.
- Thesis title: “Biofuels production from organic fraction of municipal solid waste through hydrothermal carbonization and pyrolysis” (advisors: prof. L. Fiori, prof. J. Goldfarb).
- 10/2012 – 07/2015 **Bachelor of Science in Energy Engineering**
Politecnico di Torino, Italia

FELLOWSHIP AND AWARDS

7/2024	Humboldt Research Fellowship, Alexander von Humboldt Foundation
3/2024	ACS Sustainability Star, American Chemical Society (ACS)
3/2024	International Researcher Visiting Fellowship, Tohoku University, Japan
6/2021	Best Young Researcher WasteEng conference
3/2024	Fulbright Scholarship, USA-Italy Fulbright commission
3/2024	ACS Joseph Breen Memorial Fellowship, American Chemical Society
8/2020	The Journey Fellowship, EIT-KIC Climate

DISSEMINATION AND SCIENTIFIC SERVICES

- **Public engagement:** Science outreach activities within the initiative “Women in Science” for secondary schools.
- **Science communication tools:** Development of project websites for dissemination ([ERIC SOL](#), [OcchioAlBio!](#)).
- **Peer review:** Reviewer for over 20 international scientific journals including Chemical Engineering Journal, Food chemistry, Waste Management, Nature Communication, Bioresource Technology, Fuel, Journal of Environmental Chemical Engineering.

PRESENTATIONS IN INTERNATIONAL CONFERENCES.....

2025	International Symposium on Green Chemistry – ISGC 2025. <i>La Rochelle, France.</i> May. <i>From biomass to artificial humic substances: a new path for soil health and carbon sequestration.</i>
2024	8th Expert Forum on Hydrothermal Processes. <i>Leipzig, Germany.</i> November. <i>From hydrothermal carbonization to humification for soil carbon sequestration.</i>
2024	Green Chemistry (GRS) Gordon Research Seminar. <i>Castelldefels, Spain.</i> June. <i>From biomass to artificial humic matter for soil carbon sequestration.</i>
2024	International Solvothermal and Hydrothermal Synthesis Association seminar series. Virtual. June. <i>From biomass to artificial humic matter for soil carbon sequestration.</i>
2023	27th Annual Green Chemistry & Engineering Conference. <i>Long Beach, USA.</i> June. <i>Hydrothermal carbonization as green route for valorizing bioplastic wastes.</i>
2023	3rd Symposium on Hydrothermal Carbonization. <i>Seoul, South Korea.</i> May. <i>Kinetics and chemistry behind the transition from hydrothermal carbonization to liquefaction.</i>
2022	American Chemical Society (ACS) Spring. Virtual. February. <i>From hydrothermal carbonization to liquefaction: The chemistry and kinetics along the hydrothermal biomass reaction spectrum.</i>
2021	8th International Conference on Engineering for Waste and Biomass Valorisation. Virtual. June. <i>Sun to biocoal: a solar - hydrothermal system for waste biomass valorization. Best young researcher award.</i>
2020	24th Annual Green Chemistry & Engineering Conference - Systems Inspired Design. Virtual. June. <i>Post-treatments after hydrothermal carbonization: From organic wastes to solid and liquid biofuels.</i>
2019	2nd International Symposium on Hydrothermal Carbonization, <i>Berlin, Germany.</i> May. <i>Realization of a solar hydrothermal reactor: A hybrid solution to develop a zero-energy technology.</i>

LIST OF PEER-REVIEWED PUBLICATIONS (* indicates corresponding author)

- [19] **Ischia, G.***, Marzban, N., Schmidt, J., Volikov, A. (2026). *Transitioning from hydrothermal carbonization to humification for producing artificial humic substances.* Bioresource Technology. 13336.
[\[10.1016/j.biortech.2025.133306\]](https://doi.org/10.1016/j.biortech.2025.133306)

- [18] **Ischia, G.***, Yoko, A., Wahyudiono, Fiori, L., Adschiri, T. (2026). *Valorization of lignin into carbon spheres, carbon dots, and graphene-like layers via hydrothermal conversion*. International Journal of Biological Macromolecules. 147997. [[10.1016/j.ijbiomac.2025.147997](https://doi.org/10.1016/j.ijbiomac.2025.147997)]
- [17] Zhang, Z., Xie, D., Cheng, K., Liu Z., **Ischia, G.**, Zhao, Y., Yang, F. (2025). *Optimization of Fertilization Strategies for Paddy Fields based on Multi-task Transfer Learning and Reinforcement Learning*. Agricultural Water Management. 322. 109965. [[10.1016/j.agwat.2025.109965](https://doi.org/10.1016/j.agwat.2025.109965)]
- [16] Queffelec, J., Torres Pérez, M.D., Domínguez, H., **Ischia, G.**, Filonenko, S. (2025). *Tailoring the extraction process and properties of polysaccharides from the lichen Evernia prunastri using natural eutectic solvents following a biorefinery approach*. Green Chemistry. 27, 6493–6511. [[10.1039/d5gc01413a](https://doi.org/10.1039/d5gc01413a)]
- [15] **Ischia, G.**, Marchelli, F., Bazzanella, N., Ceccato, R., Calvi M., Guella, G., Gioia, C., Fiori, L. (2024). *Cellulose Acetates in Hydrothermal Carbonization: a Green Pathway to Valorize Residual Bioplastics*. ChemSusChem, e202401163. [[10.1002/cssc.202401163](https://doi.org/10.1002/cssc.202401163)]
- [14] **Ischia, G.**, Sudibyo, H., Miotello, A., Tester J.W., Fiori, L., Goldfarb, J.L. (2024). *Identifying the Transition from Hydrothermal Carbonization to Liquefaction of Biomass in a Batch System*. ACS Sustainable Chemistry & Engineering, 12, 4539–4550. [[10.1021/acssuschemeng.3c07731](https://doi.org/10.1021/acssuschemeng.3c07731)]
- [13] **Ischia, G.**, Berge, N.D., Bae, S., Marzban, N., Román, S., Farru, G., Wilk, M., Kulli, B., Fiori, L. (2024). *Advances in Research and Technology of Hydrothermal Carbonization: Achievements and Future Directions*. Agronomy 14. [[10.3390/agronomy14050955](https://doi.org/10.3390/agronomy14050955)]
- [12] **Ischia, G.**, Goldfarb, J.L., Miotello, A., Fiori, L. (2023). *Green solvents to enhance hydrochar quality and clarify effects of secondary char*. Bioresource Technology, 129724. [[10.1016/j.biortech.2023.129724](https://doi.org/10.1016/j.biortech.2023.129724)]
- [11] **Ischia, G.**, Marchelli, F., Fiori, L. (2023). *Advancing bioenergy for a green future*. Detritus, 23, 1–2. [[10.31025/2611-4135/2023.18278](https://doi.org/10.31025/2611-4135/2023.18278)]
- [10] Marchelli, F., Ferrentino, R., **Ischia, G.**, Calvi, M., Andreottola, G., Fiori, L. (2023). *Valorisation of eyewear bioplastics through HTC and anaerobic digestion: preliminary results*. Detritus, 23, 35–42. [[10.31025/2611-4135/2023.18275](https://doi.org/10.31025/2611-4135/2023.18275)]
- [9] Marchelli, F., **Ischia, G.**, Calvi, M., Fiori, L. (2023). *Hydrothermal Carbonization as a Process to Facilitate the Disposal of Bioplastics*. Chemical Engineering Transactions, 99, 79–84. [[10.3303/CET2399014](https://doi.org/10.3303/CET2399014)]
- [8] Battocchio, P., Bazzanella, N., Orlandi, M., **Ischia, G.**, Miotello, A. (2023). *Carbon nanoparticles as absorbers in PVC for laser ablation propulsion: size effects*. Applied Physics A: Materials Science and Processing, 129, 1–12. [[10.1007/s00339-023-06502-7](https://doi.org/10.1007/s00339-023-06502-7)]
- [7] **Ischia, G.**, Cutillo, M., Guella, G., Bazzanella, N., Cazzanelli, M., Orlandi, M., Miotello, A., Fiori, L. (2022). *Hydrothermal carbonization of glucose: Secondary char properties, reaction pathways, and kinetics*. Chemical Engineering Journal, 449, 137827. [[10.1016/j.cej.2022.137827](https://doi.org/10.1016/j.cej.2022.137827)]
- [6] **Ischia, G.***, Cazzanelli, M., Fiori, L., Orlandi, M., Miotello, A. (2022): *Exothermicity of hydrothermal carbonization: Determination of heat profile and enthalpy of reaction via high-pressure differential scanning calorimetry*. Fuel, 310, 122312. [[10.1016/j.fuel.2021.122312](https://doi.org/10.1016/j.fuel.2021.122312)]
- [5] Gonnella, G., **Ischia, G.**, Fambri, L., Fiori, L. (2022). *Thermal Analysis and Kinetic Modeling of Pyrolysis and Oxidation of Hydrochars*. Energies, 15, 1–21. [[10.3390/en15030950](https://doi.org/10.3390/en15030950)]
- [4] **Ischia, G.**, Fiori, L. (2021): *Hydrothermal Carbonization of Organic Waste and Biomass. A Review on Process, Reactor, and Plant Modeling*. Waste and Biomass Valorization, 12, 2797–2824. [[10.1007/s12649-020-01255-3](https://doi.org/10.1007/s12649-020-01255-3)].
Most cited article of the journal in 2022.
- [3] **Ischia, G.**, Fiori, L., Gao, L., Goldfarb, J.L. (2021). *Valorizing municipal solid waste via integrating hydrothermal carbonization and downstream extraction for biofuel production*. Journal of Cleaner Production, 289, 125781. [[10.1016/j.jclepro.2021.125781](https://doi.org/10.1016/j.jclepro.2021.125781)]
- [2] **Ischia, G.**, Castello, D., Orlandi, M., Miotello, A., Rosendahl, L.A., Fiori, L. (2020). *Waste to biofuels through zero-energy hydrothermal solar plants: Process design*. Chemical Engineering Transactions, 80, 7–12. [[10.3303/CET2080002](https://doi.org/10.3303/CET2080002)]
- [1] **Ischia, G.**, Orlandi, M., Fendrich, M.A., Bettonte, M., Merzari, F., Miotello, A., Fiori, L. (2020). *Realization of a solar hydrothermal carbonization reactor: A zero-energy technology for waste biomass valorization*. Journal of Environmental Management, 259. [[10.1016/j.jenvman.2020.110067](https://doi.org/10.1016/j.jenvman.2020.110067)]